



Application of ELSE to FSP

An FSP reframing of the subnetwork method. ELSE is a trajectory method but the idea to resolve persistent manifolds analytically is general enough to be applied to FSP. The hypothesis is that the boundary flux the FP-FSP bounds as *error* (Thm 4.5) is exactly the method's escape density.

ELSE Overview

Subnetworks. For a subnetwork G of the full state space G_0 , edges from G to its complement are handled in one of two ways (Fig. 1 of the draft): the **recurrent (free)** subnetwork erases them (mass-conserving), and the **transient (wired)** subnetwork redirects them to a single absorbing cemetery state ω_∞ at equal total rate. Finite state projection uses the transient subnetwork; the method here also builds on the transient subnetwork and resolves its escape exactly.

Rate matrix (row convention). For rates $r(\omega_i \rightarrow \omega_j)$,

$$\mathbf{R}_{i,j} = \begin{cases} r(\omega_i \rightarrow \omega_j), & i \neq j \\ -\sum_{k \neq i} r(\omega_i \rightarrow \omega_k), & i = j, \end{cases} \quad \frac{d\vec{p}}{dt} = \mathbf{R}^\top \vec{p}, \quad \vec{p}(t) = (e^{\mathbf{R}t})^\top \vec{p}(0).$$

For the transient subnetwork the row sums are $\mathbf{R}\mathbf{1} = -\mathbf{\Delta} \leq 0$, where Δ_i is the total escape (killing) rate from ω_i ; $\mathbf{R}$ is negative definite.

Fundamental matrix (Eq. 8). The core object is

$$\mathbf{Z}_{i,j} := \int_0^\infty P_i(X_t = j) dt = \int_0^\infty \delta_i^\top e^{\mathbf{R}t} \delta_j dt = -(\mathbf{R}^{-1})_{i,j}, \quad \mathbf{Z} = -\mathbf{R}^{-1} (\geq 0),$$

the expected time spent in j before escape when started at i . From $\mathbf{Z}$ the method reads off, exactly:

- **escape-state probabilities (Eq. 10):** $p_{\text{escape}}(\omega_a) = -(\mathbf{Z}^\top \delta_a) \circ (\mathbf{R}\mathbf{1})$, normalized

since $-\mathbf{Z} \mathbf{R} \mathbf{1} = \mathbf{1}$;

- **mean escape time from a (Eq. 15):** $\mathbb{E} t_{a \rightarrow} = \delta_a^\top \mathbf{Z} \mathbf{1}$, and conditional occupation times (Eqs. 11–12);
- **occupation-time (co)variances (Eqs. 13–14)** from $\mathbf{Z}^2$;
- **cut times (Eqs. 16–18)** for pruning: via a killing-rate limit $\mathbf{R}'(\gamma) = \mathbf{R} - \gamma \delta_b \delta_b^\top$ and Sherman–Morrison, the drop in subnetwork occupation on removing state b , used to shed the least significant state.

Algorithm (exact linear system embedding). The Gillespie jump-and-hold iteration is replaced by: from the current subnetwork G , use $\mathbf{Z}$ to draw the exact escape state (Eq. 10) and the exact time to escape (Eqs. 11/15); jump there; augment G with the new state; and prune via cut time to hold $|G| \leq n_{\max}$. Escape from an entire subnetwork is collapsed into one linear solve, exactly, at cost set by $|G|$ rather than by the number of internal transitions.

Flux-Preserving FSP

Convention bridge. In `ex_article.tex` the generator is column convention, $\mathbf{A}_{ij} = \text{rate } j \rightarrow i$, $\dot{\mathbf{p}} = \mathbf{A} \mathbf{p}$. On the active set J (complement J' , block form Eq. 4.1), the transient/ wired subnetwork is the truncated generator $\mathbf{A}_{JJ}$ and the recurrent/free subnetwork is the compressed generator $\widetilde{\mathbf{A}}_{JJ} = \mathbf{A}_{JJ} + \text{diag}(\mathbf{1}^\top \mathbf{A}_{J'J})$. Fornace's rate matrix is the transpose, $\mathbf{R} = \mathbf{A}_{JJ}^\top$, so his fundamental matrix is $\mathbf{Z} = (-\mathbf{A}_{JJ}^{-1})^\top$. We work with the column-convention fundamental matrix

$$\mathbf{Z} := -\mathbf{A}_{JJ}^{-1} = \int_0^\infty e^{t\mathbf{A}_{JJ}} dt \geq 0.$$

(★)

Equation (★) is the reframing in one line: **the method's fundamental matrix is the time-integral of the FSP truncated-generator semigroup** $e^{t\mathbf{A}_{JJ}}$ — the same $\mathbf{A}_{JJ}$ whose contraction property $\|e^{t\mathbf{A}_{JJ}}\|_{1 \rightarrow 1} \leq 1$ carries Cor. 4.4 and Thm. 4.5.

FSP Error (or leakage)

The retained mass under the truncated closure is $\Gamma_J(t) = \mathbf{1}^\top e^{t\mathbf{A}_{JJ}} \mathbf{p}_0$ (survival in J). Differentiating and using the row-sum identity $\mathbf{1}^\top \mathbf{A}_{JJ} = -\mathbf{1}^\top \mathbf{A}_{J'J}$ (Eq. 4.2),

$$f(t) := -\Gamma'_J(t) = \mathbf{1}^\top \mathbf{A}_{J'J} e^{t\mathbf{A}_{JJ}} \mathbf{p}_0 = \Phi_{\text{out}}(t), \quad F(t) = \int_0^t \Phi_{\text{out}}(s) ds = 1 - \Gamma_J^{(\dagger)}(t).$$

So the **boundary outflux** Φ_{out} is the **first-passage (escape-time) density**, and the leaked mass $1 - \Gamma_J$ that FSP drives to $\leq \varepsilon$ is the **escape CDF**. The quantity the flux-adaptive method treats as truncation error to suppress (Thm. 4.5, $\|\varepsilon(t_{n+1})\|_1 \leq \int (\Phi_{\text{in}} + \Phi_{\text{out}}) ds$) is, read from the transient subnetwork, exactly what ELSE resolves as the answer.

Resolvent

Remark 4.6 already invokes the resolvent $(\lambda I - \mathbf{A})^{-1}$ through Hille–Yosida. It is the Laplace transform of the FSP semigroup,

$$(\lambda I - \mathbf{A}_{JJ})^{-1} = \int_0^\infty e^{-\lambda t} e^{t\mathbf{A}_{JJ}} dt, \quad (\ddagger)$$

where $\lambda > 0$ is the Laplace variable and the weight $e^{-\lambda t}$ down-weights late times. FSP lives in the time domain ($e^{t\mathbf{A}_{JJ}} \mathbf{p}_0$, the full transient); the method lives at $\lambda \rightarrow 0$, where the weight becomes 1 and the transform reduces to the plain time-integral $(\star)$. Whether that $\lambda = 0$ limit is finite depends on the eigenvalues of the closure, and the two FSP closures of

`ex_article.tex` differ exactly here:

- **transient / truncated** $\mathbf{A}_{JJ}$ (Cor. 4.4, strict contraction): every state escapes, all eigenvalues have $\text{Re} < 0$, so $(\ddagger)$ **stays finite at** $\lambda = 0$ and $(\star)$ gives the fundamental matrix Z ;
- **recurrent / compressed** $\widetilde{\mathbf{A}}_{JJ}$ (Prop. 4.3, stochastic semigroup): $\mathbf{1}^\top \widetilde{\mathbf{A}}_{JJ} = 0$, so 0 is an eigenvalue and $(\ddagger)$ **blows up as** $\lambda \rightarrow 0$, diverging as Π/λ along the stationary mode π .

ELSE on Truncated Generator

At $\lambda = 0$ the truncated resolvent stays finite, giving $Z = -\mathbf{A}_{JJ}^{-1}$ and, via $(\ddagger)$, the whole escape law rather than its mean. With τ the escape time (phase-type($\mathbf{p}_0, \mathbf{A}_{JJ}$)):

$$\mathbb{E}[\tau^n] = n! \mathbf{1}^\top Z^n \mathbf{p}_0, \quad \pi_{\text{exit}} = \mathbf{A}_{J'J} Z \mathbf{p}_0. \quad (1)$$

$n = 1$ is the method's mean escape time $\mathbf{1}^\top Z \mathbf{p}_0$ (Fornace Eq. 15, transposed); $n = 2$ gives $\text{Var}(\tau) = 2 \mathbf{1}^\top Z^2 \mathbf{p}_0 - (\mathbf{1}^\top Z \mathbf{p}_0)^2$ (his Eq. 14). The powers Z^n carry every moment and π_{exit} the exit-state law (his Eq. 10); the escape density itself is $f(t) = \Phi_{\text{out}}(t)$ of $(\dagger)$. The method reports the $n = 1$ slice; the resolvent supplies the full distribution.

ELSE on Compressed Generator

Motivation. FP-FSP is a simulation scheme: it truncates the infinite-dimensional generator $\mathbf{A}$ into a sequence of finite matrices $\mathbf{A}_{J_n J_n}$, evolves each over a window $[t_n, t_{n+1}]$, and stitches the evolutions together. Per window the transient closure $\mathbf{A}_{J_n J_n}$ answers one question: how mass leaves J_n . Its mass drains, so the only surviving content is the escape law above. The compressed closure $\widetilde{\mathbf{A}}_{J_n J_n}$ instead reflects escaping mass back, giving a mass-conserving chain on the window with a stationary distribution $\pi^{(n)}$ (unique when J_n is connected, which the flux pruning preserves) and the quantities leakage destroys: within-window mean first-passage times, and how fast a time-average $\frac{1}{T} \int_0^T g(X_t) dt$ settles. This is the recurrent (free) versus transient (wired) subnetwork distinction of nuts.pdf, where the transient/FSP closure "destroys many averaged quantities" while the free closure preserves them.

The $\pi^{(n)}$ is local in time: the equilibrium of the reflected dynamics on the current window, not a truncation of any global CME stationary law (which may sit at infinity or fail to exist). For a spreading distribution it is partly an artifact of the reflecting wall, but on a window that captures a metastable basin it is the within-basin quasi-equilibrium and D its relaxation structure. Worth exploring.

Resolvent. Write $\widetilde{\mathbf{A}} := \widetilde{\mathbf{A}}_{JJ}$, with $\widetilde{\mathbf{A}}\pi = \mathbf{0}$ and $\mathbf{1}^\top \pi = 1$. Let $\Pi := \pi \mathbf{1}^\top$; for any distribution $\mathbf{v}$, $\Pi \mathbf{v} = (\mathbf{1}^\top \mathbf{v}) \pi = \pi$, so Π collapses everything onto π . It is the rank-one matrix whose every column is π (outer product of the column π with the row $\mathbf{1}^\top$), equal to the infinite-time limit $\lim_{t \rightarrow \infty} e^{t\widetilde{\mathbf{A}}}$ and to the eigenprojector for the zero eigenvalue of $\widetilde{\mathbf{A}}$. Two facts drive the rest:

$$\widetilde{\mathbf{A}} \Pi = (\widetilde{\mathbf{A}} \pi) \mathbf{1}^\top = \mathbf{0}, \quad \Pi \widetilde{\mathbf{A}} = \pi (\mathbf{1}^\top \widetilde{\mathbf{A}}) = \mathbf{0},$$

i.e. Π singles out the zero-eigenvalue direction and $I - \Pi$ is everything else, where $\widetilde{\mathbf{A}}$ has all eigenvalues with $\text{Re} < 0$. Splitting the semigroup along this decomposition,

$$e^{t\tilde{\mathbf{A}}} = \Pi + \underbrace{e^{t\tilde{\mathbf{A}}}(I - \Pi)}_{\rightarrow 0 \text{ exponentially}},$$

which is just $e^{t\tilde{\mathbf{A}}}\mathbf{p}_0 \rightarrow \boldsymbol{\pi}$ (the equilibration of Prop. 4.3) restated. Feed this into the resolvent integral:

$$(\lambda I - \tilde{\mathbf{A}})^{-1} = \int_0^\infty e^{-\lambda t} e^{t\tilde{\mathbf{A}}} dt = \Pi \int_0^\infty e^{-\lambda t} dt + \int_0^\infty e^{-\lambda t} (e^{t\tilde{\mathbf{A}}} - \Pi) dt.$$

The first integral is Π/λ and carries the entire blow-up; the second stays finite at $\lambda = 0$ because $e^{t\tilde{\mathbf{A}}} - \Pi$ decays. Evaluating it at $\lambda = 0$ defines the remainder:

$$D := \int_0^\infty (e^{t\tilde{\mathbf{A}}} - \Pi) dt, \quad \text{giving} \quad (\lambda I - \tilde{\mathbf{A}})^{-1} = \frac{\Pi}{\lambda} + D + O(\lambda). \quad (2)$$

This is the exact recurrent mirror of $(\star)$: there we integrated $e^{t\mathbf{A}_{JJ}}$ directly because it decayed to $\mathbf{0}$; here $e^{t\tilde{\mathbf{A}}}$ decays only to $\boldsymbol{\pi}$, so we subtract $\boldsymbol{\pi}$ first and integrate the remainder. D (the *deviation matrix*) is the accumulated gap between the chain and its stationary law, and satisfies $-\tilde{\mathbf{A}} D = I - \Pi$: it is $\tilde{\mathbf{A}}$ inverted everywhere except along the stationary direction.

Quantities. From D one reads the long-time quantities that leakage would have destroyed: the within- J mean first-passage times $m_{xy} = (D_{yy} - D_{xy})/\pi_y$ (the recurrent analog of the mean escape time), and how fast a time-average $\frac{1}{T} \int_0^T g(X_t) dt$ settles onto its stationary value $\sum_x \pi_x g(\mathbf{x})$.

Next steps. Demonstrate the escape density $f(t) = \Phi_{\text{out}}(t)$ and variance (1) on the fast/slow example (`examples/fast_slow_else.jl`) against SSA escape-time histograms — the mean is one number off a distribution the resolvent gives in full.